

M. G. SAI SWARUP

Software Engineer | Python, Systems & Application Developer

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ABOUT ME

Curious and practical-minded Computer Applications graduate with a strong passion for software development, system architecture, and automation. I enjoy building clean, functional tools that solve real-world problems—spanning database-driven web applications, real-time sync systems, desktop GUIs, and Linux network scripts. Known for proactive problem solving, quick adaptability, and a structured approach to writing reliable code and debugging systems under varied network environments.

TECHNICAL SKILLS & TOOLKIT

- **Languages & Database:** Python, JavaScript, SQL, HTML5, CSS3, Bash/Shell Scripting
- **Frameworks & Web:** Flask, RESTful APIs, WebSockets, Tkinter (Desktop GUI)
- **Tools & Platforms:** Git & GitHub, Postman, VirtualBox, VS Code, Linux (Kali / Ubuntu / Termux)
- **Core Concepts:** Custom Cryptographic Logic, Real-Time Sync Protocols, Automation Scripting, STLC / Testing

FEATURED PROJECTS

E-Commerce Platform (Shadow Market) [\[Python, JavaScript, SQL, HTML5/CSS3\]](#) Deployed Web App

- Engineered and deployed a responsive online store featuring product catalog browsing, cart management, and user authentication.
- **Tools & Platforms:**

ShadowPad — Collaborative Workspace & Custom Ratchet Sync [\[JavaScript, WebSockets, Cryptography, HTML5/CSS3\]](#) Completed Project

- Designed and implemented the custom 'Shadow Ratchet' algorithm (inspired by the Double Ratchet Algorithm) for secure, forward-secret session key derivation and real-time state synchronization.
- Developed a low-latency collaborative rich-text editor over WebSockets, preventing concurrent race conditions and message replay.

Flask Student Management System [\[Python, Flask, SQLite/MySQL, HTML/CSS\]](#) Completed Project

- Built a full database records manager supporting complete student CRUD operations, grading workflows, and secure access.
- Implemented structured error handling, form sanitization, and optimized queries for zero-crash execution.

GUI Code Assistant & Automation Bot Suite [\[Python, Tkinter, Automation Scripts\]](#) Completed Tools

- Constructed a desktop developer utility application for code shortcuts, syntax formatting, and reference lookups.
- Scripted automated background processes and uptime polling bots for scheduled workflow management.

ShadowProbe — Security & Network Auditor [\[Python, Bash, Kali Linux, VirtualBox\]](#) In Active Development

- Building an automated auditing tool to scan subnets, inspect open ports, and detect host configuration vulnerabilities.
- Testing network socket handling and modular reporting inside virtualized Linux test environments.

CERTIFICATIONS & CREDENTIALS

- **n8n: No Code AI Agent Builder:** Simplilearn Skillup (Code: 9913759)
- **Python Essentials 2:** Cisco Networking Academy
- **Software Testing Internship:** Experience Verification
- **AWS / Institutional Certification:** VFSTR (ID: 231FJ01042)
- **Automated Testing:** QA Advanced Certification
- **Security Workshop Merit:** Vignan University Workshop

EDUCATION

- **Bachelor of Computer Applications (BCA)** — Vignan University (VFSTR), Vadlamudi (2023 – 2026)
- **Intermediate (BiPC)** — NRI Junior College, Andhra Pradesh (2021 – 2023)
- **Secondary School Certificate (SSC / 10th)** — Dr. KKR's Gowtham School, Andhra Pradesh (Completed 2021)